FRC SENSOR CAN — REVISION C LITE PROTOTYPE

Three fixed-threshold digital inputs. 6–22 V input; protected battery and regulated 5 V sensor power.
STM32F042 + TCAN3413. Digital status and edge counts over a custom team CAN protocol.
Not a CTRE Phoenix-compatible device. Inspect and bench-test prototypes before robot use.

Battery entry and internal supplies

File: power.kicad_sch

Protected sensor power rails

File: sensor_power.kicad_sch

Controller, clock, debug and status

File: controller.kicad_sch

CAN physical interface and termination

File: can.kicad_sch

Input 1: fixed-threshold digital input

File: input1.kicad_sch

Input 2: fixed-threshold digital input

File: input2.kicad_sch

Input 3: fixed-threshold digital input

File: input3.kicad_sch

Original design — custom WPILib CAN protocol

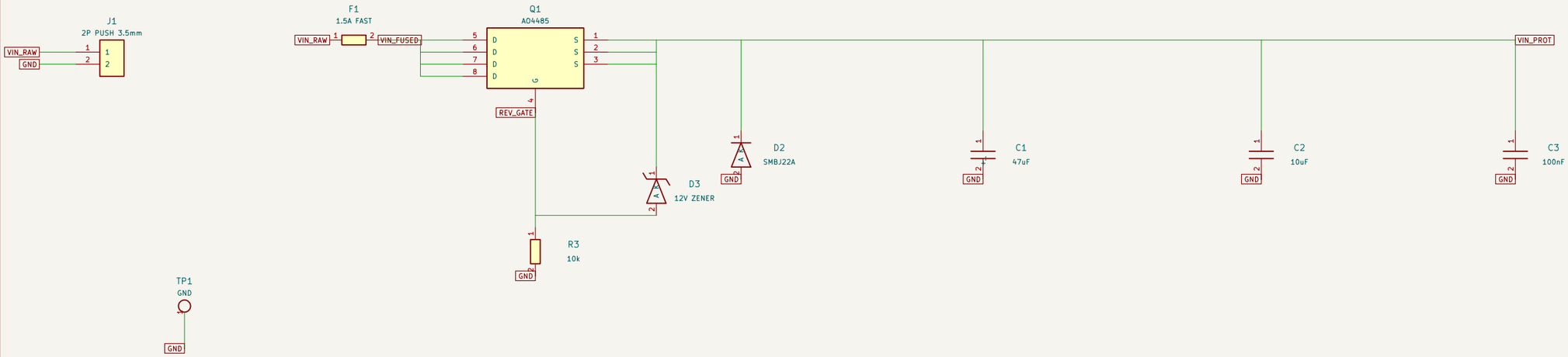
Sheet: /
File: FRC_Sensor_CAN.kicad_sch

Title: FRC Sensor CAN

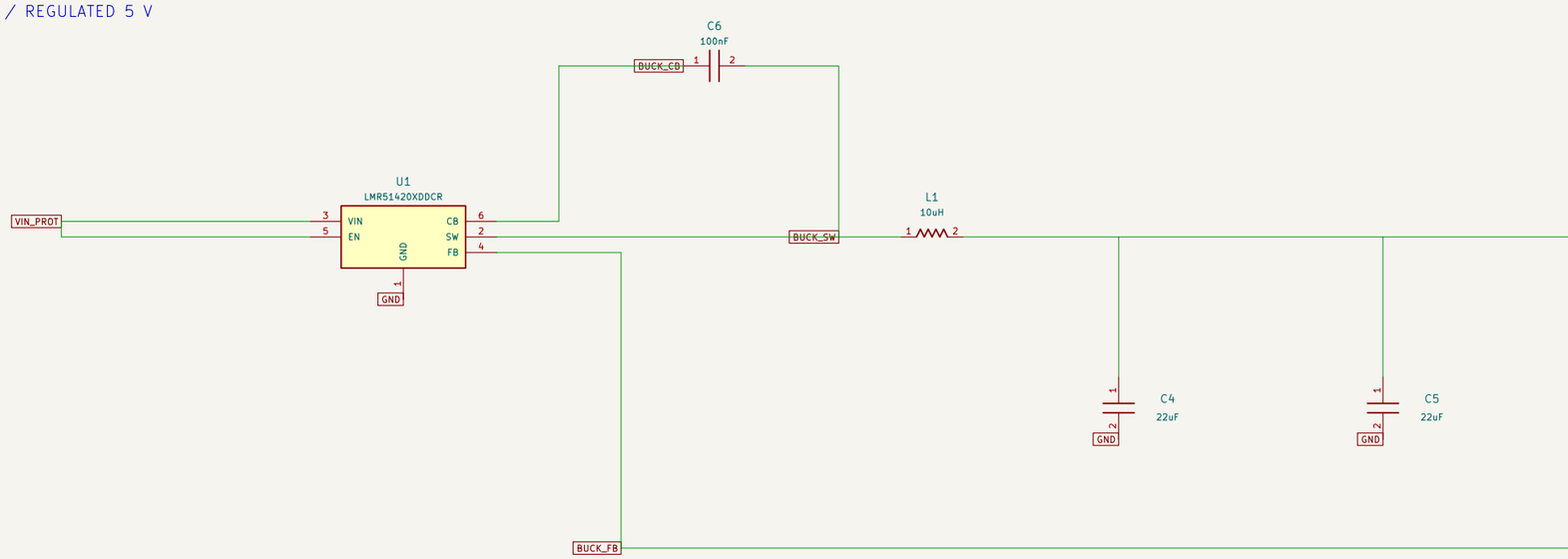
Size: A3	Date: 2026-09-08	**Rev: C LITE PROTOTYPE**
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Battery entry and internal supplies

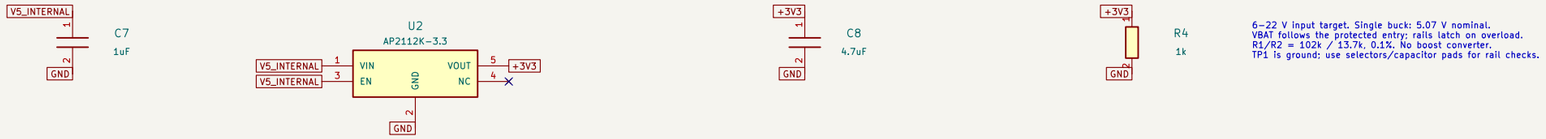
PROTECTED BATTERY ENTRY



500 kHz BUCK / REGULATED 5 V



3.3 V LOGIC SUPPLY



6–22 V input target. Single buck: 5.07 V nominal.
VBAT follows the protected entry; rails latch on overload.
R1/R2 = 102k / 13.7k, 0.1%. No boost converter.
TP1 is ground; use selectors/capacitor pads for rail checks.

Prototype: bench validation required before robot use

Original sensor interface design

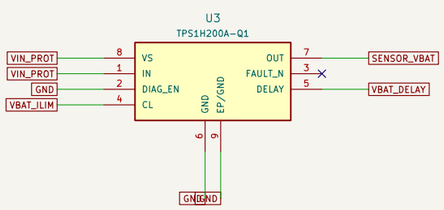
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File: power.kicad_sch

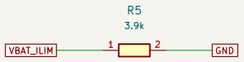
Title: FRC Sensor CAN — Battery entry and internal supplies

Size: A3	Date: 2026-09-08	**Rev: C LITE PROTOTYPE**
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Protected sensor power rails

HARDWARE CURRENT LIMITING; NO FIRMWARE ENABLE



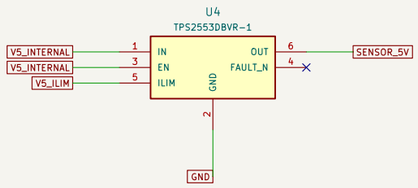


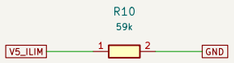






Battery output: U3 IN tied to VS; DIAG_EN grounded, FAULT unconnected.
TI supports this standalone connection. 3.9k sets the current limit; 33nF sets latch delay.
Cycle board power after correcting an overload. Powered sensor ports remain live during MCU reset.









Regulated output: TPS2553-1 latch-off version, EN tied to its input.
59k current-limit resistor. Cycle board power to clear overload.
Design target: 300mA combined sensor load per rail, not per connector.
Do not back-power the sensor supply outputs.

Prototype: bench validation required before robot use
Original sensor interface design

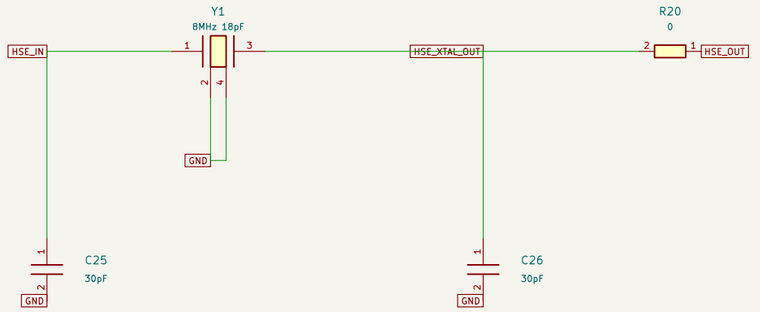
Sheet: /Protected sensor power rails/
File: sensor_power.kicad_sch

Title: FRC Sensor CAN – Protected sensor power rails

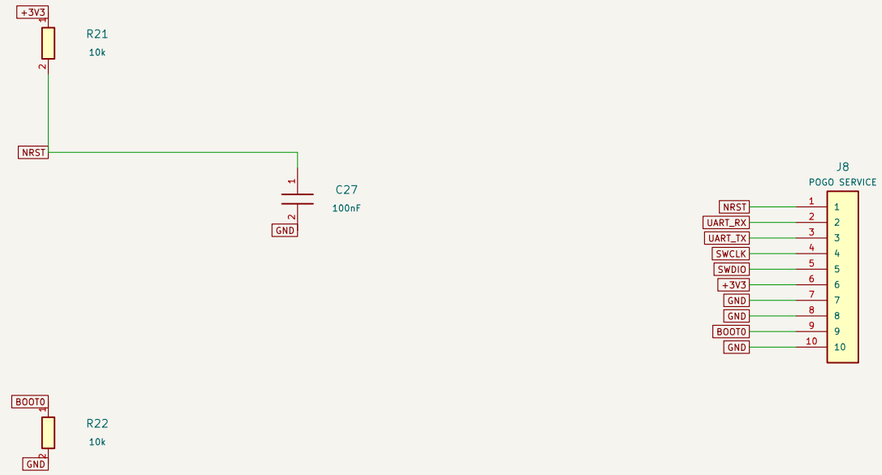
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Controller, clock, debug and status

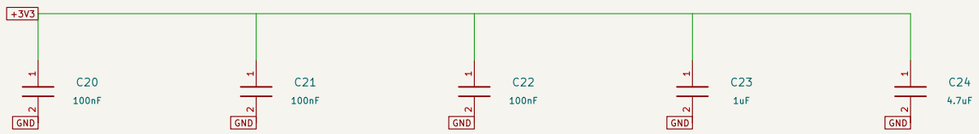
8 MHz CRYSTAL / PLL x6

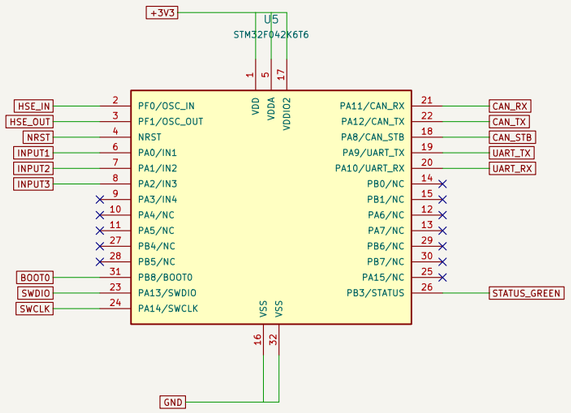


RESET / BOOT / SERVICE



LOCAL DECOUPLING





TC2050-IDC-NL mechanical footprint; CUSTOM electrical pinout.
Pin 6 is 3.3 V reference OUTPUT, never a programming power input.
Unused GPIOs are unconnected and initialized to analog mode.
No rail ADCs, software enable signals or per-input LEDs.

Prototype: bench validation required before robot use

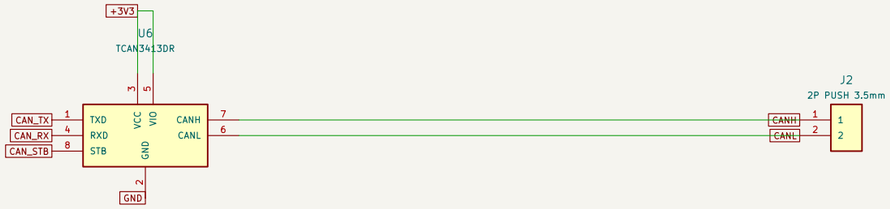
Original sensor interface design

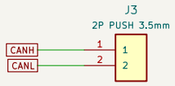
Sheet: /Controller, clock, debug and status/
File: controller.kicad_sch

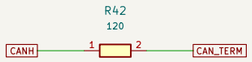
Title: FRC Sensor CAN – Controller, clock, debug and status

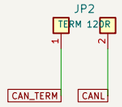
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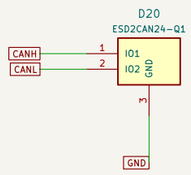
CAN physical interface and termination

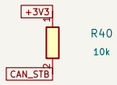


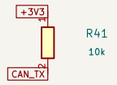




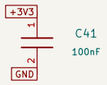






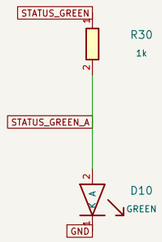






1 Mbit/s. Bridge JP2 only at a physical CAN trunk end.
TCAN3413 uses 3.3 V for both VCC and VIO.
STB pull-up keeps the bus silent during MCU reset.
CAN ground is shared through the power connection.

SINGLE STATUS INDICATOR

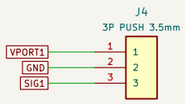


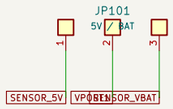
Prototype: bench validation required before robot use
Original sensor interface design

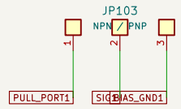
Sheet: /CAN physical interface and termination/		
File: can.kicad_sch		
Title: FRC Sensor CAN — CAN physical interface and termination		
Size: A3	Date: 2026-09-08	**Rev: C LITE PROTOTYPE**
KiCad E.D.A. 10.0.6		Id: 5/8

Input 1: fixed-threshold digital input

SOLDER CONFIGURATION: SELECT POWER AND SENSOR OUTPUT TYPE





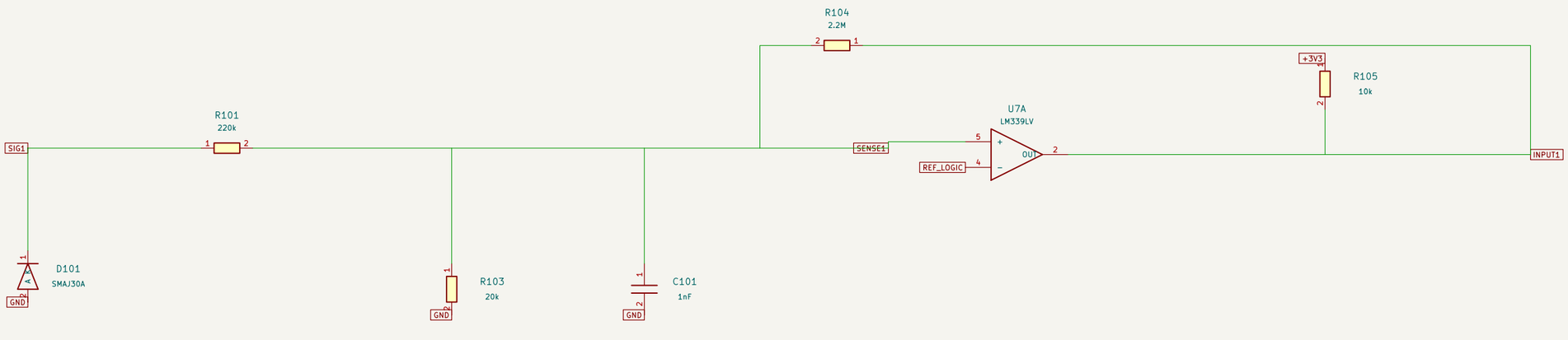


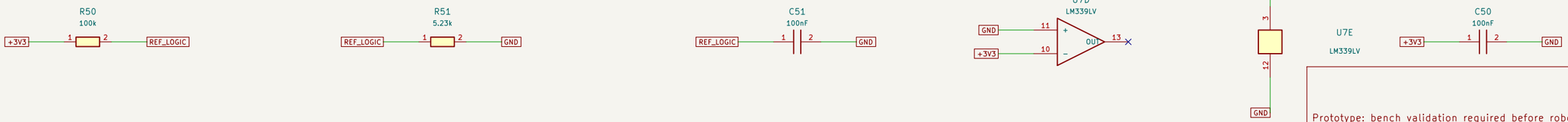






POWER: bridge 1-2 for 5 V OR 2-3 for battery. NEVER bridge all three.
BIAS: bridge 1-2 for NPN, 2-3 for PNP, leave open for push-pull/external bias.
All selectors are open as fabricated. Voltage-tolerant NPN output required for selected pullup.





0-30 V signal design range; software debounce follows the hardware filter.
Nominal thresholds near 2.0 V rising / 1.65 V falling. Final bounds are in electrical_checks.json.
No analog measurement, 1.8 V logic guarantee or CTRE protocol compatibility.

Prototype: bench validation required before robot use
Original sensor interface design

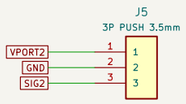
Sheet: /Input 1: fixed-threshold digital input/
File: input1.kicad_sch

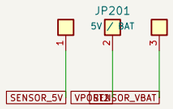
Title: FRC Sensor CAN — Input 1: fixed-threshold digital input

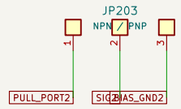
Size: A3	Date: 2026-09-08	**Rev: C LITE PROTOTYPE**
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Input 2: fixed-threshold digital input

SOLDER CONFIGURATION: SELECT POWER AND SENSOR OUTPUT TYPE





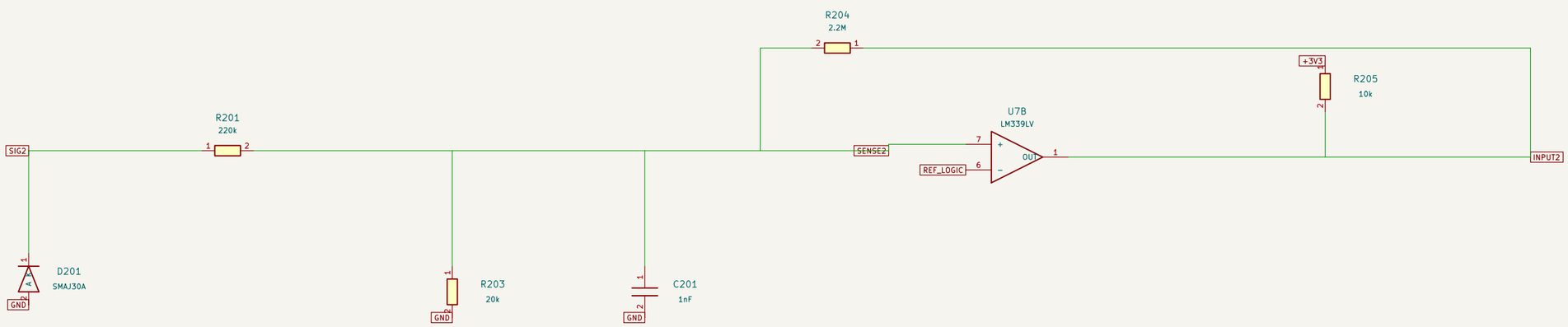








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The reference divider/filter and comparator power connections are on input1.
One fixed logic-compatible threshold for all three channels; no industrial threshold selector.

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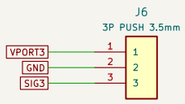
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File: input2.kicad_sch

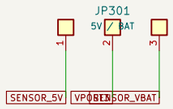
Title: FRC Sensor CAN - Input 2: fixed-threshold digital input

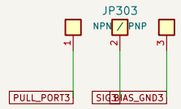
Size: A3	Date: 2026-09-08	**Rev: C LITE PROTOTYPE**
KiCad E.D.A. 10.0.6		Id: 7/8

Input 3: fixed-threshold digital input

SOLDER CONFIGURATION: SELECT POWER AND SENSOR OUTPUT TYPE

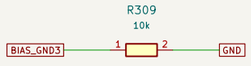




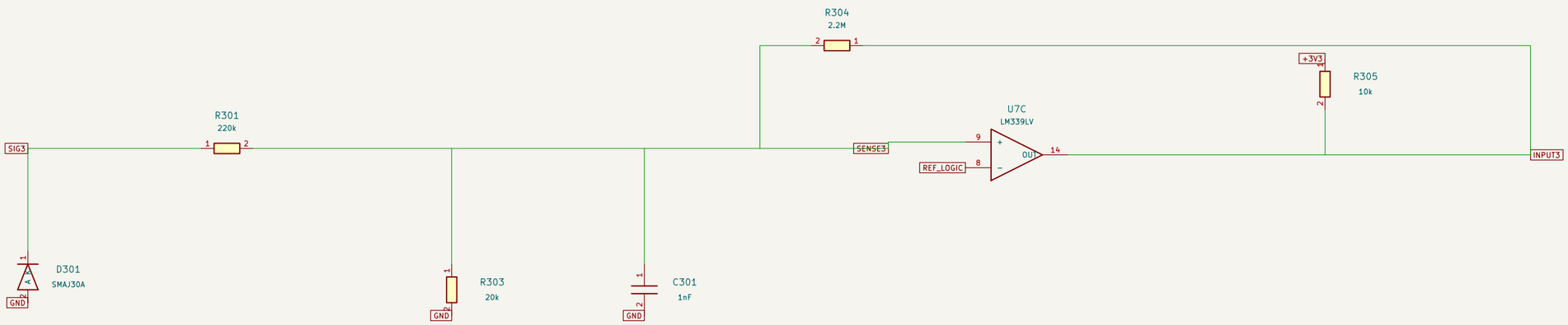








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Original sensor interface design

Sheet: /Input 3: fixed-threshold digital input/
File: input3.kicad_sch

Title: FRC Sensor CAN – Input 3: fixed-threshold digital input

Size: A3	Date: 2026-09-08	**Rev: C LITE PROTOTYPE**
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